

Linear Algebra Final Exam 2024-02-07

Use a separate answer sheet for each problem. Make sure to write your name, student ID and the problem number on each sheet.

1. (a) Find all the solutions of $A\mathbf{x} = \begin{bmatrix} 1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 12$.

(b) Find all the solutions of $A\mathbf{x} = \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ -4 \end{bmatrix}$.

2. Find the number of independent vectors of these 4 spaces.
Which two of the spaces are the same?

(a) Column space of A , (b) column space of U , (c) row space of A , (d) row space of U :

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

3. Project $\mathbf{b}$ onto the column space of A by solving $A^T A \mathbf{x} = A^T \mathbf{b}$ ($\mathbf{p} = A\mathbf{x}$):

(a) $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$

4.

- (a) Write down three equations for the line $b = C + Dt$ to go through $b = 1$ at $t = -1$, $b = 2$ at $t = 0$, and $b = 3$ at $t = 1$. Find the least squares solution $\mathbf{x} = (C, D)$.

- (b) Find $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ (orthonormal) as combinations of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ (independent columns).
Then write A as QR :

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

Overall score:

Average of quiz (AQ): remove the highest two and lowest one.

$$\text{AQ} * 0.30 * 0.87 + \text{midterm} * 0.35 + \text{finalterm} * 0.35$$

Answers

1. (a) Find all the solutions of $Ax = \begin{bmatrix} 1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

- (b) Find all the solutions of $Ax = \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ -4 \end{bmatrix}$.

(a) $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow [A, b] \rightarrow R = \begin{bmatrix} 1 & 0 & -1 & 8 \\ 0 & 1 & 1 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

2. Find the number of independent vectors (dimensions) of these 4 spaces.

Which two of the spaces are the same?

(a) Column space of A , (b) column space of U , (c) row space of A , (d) row space of U :

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

All the dimensions are 2, (c) and (d) are the same

3. Project b onto the column space of A by solving $A^T A \hat{x} = A^T b$ and $p = A\hat{x}$:

(a) $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$

(a) $A^T A \hat{x} = A^T b$ $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \hat{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \rightarrow \hat{x} = \begin{bmatrix} -2 \\ 3 \end{bmatrix} \rightarrow p = A\hat{x} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$

(b) $A^T A \hat{x} = A^T b$ $\begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} \hat{x} = \begin{bmatrix} 8 \\ 14 \end{bmatrix} \rightarrow \hat{x} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} \rightarrow p = A\hat{x} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$

4. (a) Write down three equations for the line $b = C + Dt$ to go through $b = 1$ at $t = -1$, $b = 1$ at $t = 0$, and $b = 2$ at $t = 1$. Find the least squares solution $\hat{x} = (C, D)$.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad A^T A \hat{x} = A^T b \quad \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \hat{x} = \begin{bmatrix} 6 \\ 2 \end{bmatrix} \rightarrow \hat{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- (b) Find q_1, q_2, q_3 (orthonormal) as combinations of a, b, c (independent columns). Then write A as QR :

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = q_1$$

$$u_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \rightarrow q_2 = \sqrt{\frac{1}{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \rightarrow q_3 = \sqrt{\frac{1}{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix} = [q_1, q_2, q_3] \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2\sqrt{2} & 3\sqrt{1/2} \\ 0 & 0 & \sqrt{1/2} \end{bmatrix}$$